Orange County Fire Services
Operational Area Plan Annex

Section 5
PLANS

Swift Water Rescue Plan

Prepared by:
Orange County Fire Chiefs' Association
Orange County Fire Services
Operational Plan

INTENT

The purpose of this plan is to provide standard operating procedures for the following situations in Orange County:

1. Individual water rescue incidents occurring in rivers, streams, flood control channels, aqueducts, and flash flood zones.

2. Rescue activities in response to imminent/actual flooding and dam failure.

3. Mud and Debris Flow – post wildland fire events

This document applies to all uniformed personnel including:

- Orange County Fire Personnel
- Orange County Law Enforcement
- Orange County Lifeguards

Due to the strategic geographic location of Fire Stations, the Fire Department has primary responsibility for most Swiftwater/Flood Search and Rescue (SF/S&R) Operations in Orange County. Since SF/S&R emergencies involve both *static* and *dynamic* situations, rescuers should stress the ability to remain flexible in their rescue attempts. Recognized Orange County SF/S&R teams, reference Appendix C, shall utilize FIRESCOPE ICS-SF/S&R 020-1 Swiftwater/Flood Search and Rescue, *"Recommended Training, Skills and Equipment List"* and FIRESCOPE, ICS US&R 120-2 Swiftwater/Flood Search and Rescue "*Operational System Description & Law Enforcement Mutual Aid Plan (SAR) Annex"*

PROCEDURE

A. Company Officers

   1. The first unit onscene should respond to the point where the victim(s) was last seen. "Point Last Seen".
   2. The additional responding companies report to pre-determined downstream rescue locations (Bridges) as directed by the IC.
   3. Report information to the Incident Commander (IC).
   4. Set-up for "*Low Risk – Shore Based Rescue Operations"*, reference Appendix A.

B. Incident Commanders

   1. Set-up a single Command Post at any convenient location and assume Command of the Incident.
   2. Coordinate the deployment of units to pre-determined rescue locations as per

this plan, including downstream agencies. Incident Command shall reside with the IC from the initiating agency and include all downstream agencies to the incidents geographical termination point.

3. Have the nearest Fire or Regional Law Enforcement helicopter begin searching the involved waterway as soon as possible.
4. Divide the incident into Divisions with a separate Division for each Rescue location.
5. Take necessary action to prevent loss of life in a manner consistent with this document and your Department Policy.

C. Law Enforcement and/or Lifeguards may be utilized for assistance with searches, and dive rescue team responses.

D. Army Corps of Engineers shall notify the Area Coordinator, Orange County Fire Authority, who in turn will broadcast this information over the Dispatch-to-Dispatch Channel (DSP-DSP), so that all Dispatch Agencies can be put on Alert Status when extra water is expected to be released from dams affecting Orange County.

E. Incident Priorities when dealing with any water rescue incident shall be those listed below:

1. The Protection of Life – First Responders First, Victim(s) Second.
2. Provide Search, Rescue, and assistance to survivors.
3. Provide notification and assistance to other agencies (Law Enforcement, Lifeguards, etc.)
4. Scene control as required to limit access within 10 feet of the *Water's Edge* to only those personnel who are *Properly Trained* and wearing the recognized water safety equipment; reference Appendix C.

F. Command:

Because these incidents involve moving water, the incident shall be broken down into Divisions, with each Division covering a Rescue Location and the geographical area between that Rescue Location and the next Downstream Rescue Location. For Example:

1. Divisions will be identified by rescue location, typically using bridges.
2. The Entry Point division shall be identified by the closest upstream bridge and shall geographically consist of the area from the Entry Point to the next downstream bridge/Division.
3. Subsequent Divisions shall geographically consist of the area from the bridge/Division downstream to the next bridge/Division. For example, Katella Division consists of the Katella Bridge and the area downstream to the

(upstream side of) the Orangewood Bridge.

4. Where bridges are not available and/or used as Division breaks, the same concept applies regardless of the geographical landmark used.
5. Based on water conditions and status of the victim (moving or static rescue), the IC has the discretion to scale thew incident by staffing only the necessary downstream bridges. Additionally, based on condition reports from downstream Divisions, if little to no water is moving in their respective division, the IC can dissolve remaining downstream divisions.

G. Foot Search:

After downstream units are in place and as soon as Victim(s) Information has been passed on to the Incident Commander, the First-In Company should begin a Foot Search. This Primary Search should follow the banks of the channel in the direction of the flow of water. Focus on potential entrapment hazards (i.e., low head dams, debris piles, vegetations, etc.). When the First Search Team reaches the First Rescue Location, the crews assigned to that Rescue Location will continue the Primary Search in their Division until they reach the Second Rescue Location, where the next team will continue the search, etc. A Secondary Search should be conducted if the Primary Search fails to locate the victim(s).

H. Positions:

1. Based on incident complexity, consider the use of expanded ICS positions to maintain span of control.
   a) BRANCH – could be utilized in a full-scale response to provide better span of control
   b) RESCUE GROUP SUPERVISOR – could be utilized to coordinate technician level, water entry teams. An RGS would also be appropriate for a static water rescue operation where multiple entry teams are working together to perform the rescue.

I. Resources

1. The following resources can be utilized to assist in the event of a confirmed Swiftwater rescue in the Santa Ana River.
   a) OC Sheriff
   b) Flood Control
   c) Army Corp of Engineers
   d) Law enforcement Helicopter with F.L.I.R.
   e) Recue dive team
   f) Department UAS (unmanned aerial surveillance)

Tactical Priorities

A. Standard Response

1. A SF/S&R operation will require multiple units set up at multiple rescue locations to ensure the safety of rescuers and increase the likelihood of a successful rescue.
2. The number of units required at each Rescue Location will vary based on the conditions at that location.
3. SF/S&R Personnel should Pre-plan each potential Rescue Location to best determine how many units would be required at each location.
4. Once the required number of units is determined for each Pre-planned Rescue Location, these units should be added as part of an automatic dispatching for this Rescue Site for a Swiftwater Rescue Incident.

B. Initial Size Up

The initial Size-Up is particularly important when dealing with moving water. Conditions can vary drastically depending on the volume and speed of water, weather, time of day, etc. The Size-Up report should include:

1. Number and description of victim(s), and “Point Last Seen”.
2. Conditions at “Point Last Seen” (water depth, speed, etc).
3. Assessment of hazards to personnel (debris, low head dams, etc).
4. Initial action and instructions to responding units. This shall be based on any
preplans for the waterway and conditions at scene.

C. Rescue

The Incident Commander should get a report by Divisions to make sure that units are ready to conduct immediate rescue operations if the victim(s) is found.

1. Units should stand by with rescue systems ready to deploy until otherwise directed by the Incident Commander.

2. Any unit spotting the victim(s) shall notify the Incident Commander with their current location and attempt a rescue.

    a. If immediate rescue is not possible, personnel shall try to maintain visual contact with the victim and track his/her location as long as possible. Attempt to aid victim with a flotation device.
    b. Notify downstream units to prepare for rescue attempts.

3. The Division Supervisors will coordinate the rescue effort in their Divisions, and should verify that multiple downstream rescue methods and safety's are in place, in the event that a rescuer or a victim(s) are swept further downstream.

    a. Provide Medical Aid as needed.
    b. The Incident Commander is responsible for termination of search and rescue operations.
    c. Search operations are to continue as long as there is reason to suspect that victim(s) may still be alive.
    d. Demobilization: Normally, upstream units are released first, followed by downstream units in descending order. The Incident Commander can begin releasing units as soon as he is reasonably sure that the victim has passed their locations and there are no further assignments.

Communications

The goal is to have the predetermined number of Rescue Units waiting at each Rescue Site downstream of where the victim entered the water. With this in mind, the following will be the communications and notification protocol for all responding units and their jurisdictional dispatch centers:

A. After dispatching units from your jurisdiction to the reported point of entry, and predetermined rescue locations, immediately notify "*All Applicable Agency Dispatch Centers"* that a "Water Rescue Incident" is in progress. The dispatch centers should immediately coordinate with the first responding Battalion Chief to ensure the Point of Entry, and the next 3 bridges/overpasses are staffed with rescue personnel. It will be at the discretion of the IC to staff additional downstream divisions based on the needs of the incident. The Battalion Chief should, through the appropriate dispatch center, request Law Enforcement response to rescue locations in order to provide traffic control.

B. The nearest Fire or Regional Law Enforcement Helicopter should be requested to begin an *"Aerial Search"* of the waterway. Helicopters should communicate pertinent information with the Incident Commander on the assigned Command Channel or designated Air-to-ground frequency.

C. Command Channel:

The jurisdictional dispatch center should utilize a County Wide Communications Channel and ensure ALL responding units are dispatched on the command channel. The command channel is primarily used for communications as follows:

- IC to Division
- Division to Division
- IC to Helicopter (5A will be utilized for Air-to-Ground communications with OCFA helicopters.

Primary Command Frequency	Alternate Command Frequency
2L	**3L**

*frequency will be assigned by the originating dispatch center. If the predesignated frequencies are unavailable, an alternate frequency shall be assigned to the incident and made known to all responding units.

D. Tactical Channel:

All responding units should continually monitor their normal agency tactical channels in addition to the incident command channel. The Tactical Channel is used for:

1. Communication within each division
2. Maintain jurisdictional awareness

ANA	ORG	ORC	FVY	COS	HTB	NPB
2F	**2I**	**4F**	**3G**	**3H**	**3H**	**3I**

*frequency will be assigned by the originating dispatch center. If the predesignated frequencies are unavailable, an alternate frequency shall be assigned to the incident and made known to all responding units.

E. Talk Around Channel:

Rescue teams established within a division may also use a talk around channel for team-to-team communications.

Annual Training

A Swiftwater rescue is a complex incident that has the potential to span multi-jurisdictional boundaries. The operational area should conduct countywide Swiftwater Rescue training each year (to take place in the 4th quarter after fire season and prior to the start of the rainy season). This training should cover the currently adopted OC Annex Swiftwater Rescue Plan, communications, command & control, risk assessment, and shore-based rescue operations. The training will be hosted by each participating agency on a rotational basis.

Santa Ana River Pre-Plan

A WUI style pre-plan should be created to identify all overpasses, access points, and hazards along the Santa Ana River. The plan should cover the area from Green River to W. Coast HWY. It should be simple and utilize a QR code to be made readily available to responding units.

APPENDIX A

SWIFTWATER/FLOOD SEARCH AND RESCUE PROCEDURES

Rescuers assigned to a SF/S&R Incident should take steps to make sure that the following guidelines are followed:

A. First Responders should begin to implement "Low Risk – Shore Based Operations" immediately upon arriving onscene. "High Risk – In Water Contact Operations" should be limited to properly trained, equipped and recognized SF/S&R Personnel.

B. All personnel assigned to the incident should make sure that they are prepared to:

   1. Rescue themselves first.
   2. Back each other up with multiple downstream safety's in place.
   3. After 1 and 2 are accomplished, then start setting up for the rescue of the victim(s).

C. There are several types of acceptable rescue methods which are listed below from Low Risk to High Risk, and these methods are further broken down into two rescue categories:

   - Low Risk– Shore Based Rescues (SF/S&R Awareness Level First Responders)
   - High Risk– In-Water Contact Rescues (SF/SAR Teams), reference Appendix D for recognized Orange County SF/S&R Teams.

   1. Low Risk – Shore Based Rescues (First Responders)

      a. TALK:
         Try to coach the victim through a self-rescue.

      b. PROVIDE FLOTATION:
         Give victim(s) some sort of flotation (PFD, ring buoy, Etc.)

      c. REACH:
         Reach the victim from shore using a pike pole, ladder, etc.

      d. THROW:
         Throw a Throw-bag with a rope attached and pendulum the victim to the shore.

APPENDIX A (Continued)

e. INFLATED FIRE HOSE FROM A BRIDGE:
Inflate a section of 2.5" hose, (no longer than 100'), and secure with rope on each end to lower into the water. Once victim(s) make contact with the hose, pendulum the victim to the shore (length of selected hose is incident dependent).

f. TENSION DIAGONAL LINE:
By using a rope long enough to reach across the waterway and fastened on each side at an angle to the current (45 to 60 degrees); you can use the force of the water to push the victim(s) or rescuer to the side of the channel where the downstream end of the rope is.

2. High Risk – In Water Contact Rescues (SF/S&R Teams)

a. ROW:
Utilize a Tethered Boat Systems.

b. GO and TOW:
Swim out to the victim(s) and make a contact rescue.

c. BOAT, POWERED:
Personal Watercraft (PWC), Inflatable Rescue Boat (IRB), Rigid Hull Inflatable Rescue Boat (RHIRB)

d. HELICOPTER:
The primary role of helicopters in a SF/S&R Incidents will be for aerial search to locate and track victim(s). Helicopters may be used in an actual SF/SAR Operations only with personnel trained and qualified with Helicopter/Aquatic Rescue Operational Skills. Helicopters being used for In-Water Contact Live Rescues should comply with FIRESCOPE ICS-US&R-120-2, Page 10, Appendix D – Air Resource Typing.

When reporting of GPS coordinates with Air Resources, it is vital to give your waypoints in the same Map Datum and latitude and longitude coordinates that is being utilized by the Air Resources. The preferred Map Datum to be used is NAD 83 (CONUS). The correct standard to express latitude and longitude coordinates in will be *"Degrees: Minutes.Decimal Minutes" (dd:mm.mm).*

APPENDIX B

MUD AND DEBRIS FLOW PROCEDURES

Incidents involving mud and debris flow present a special concern to water rescue personnel. Typically, these events occur after a significant wildland fire. The definition of a Mud and Debris Flow is: *"A moving mass of loose mud, sand, soil, rock, debris, water and air that travels down a slope under the influence of gravity".* In areas of very steep slopes, they can reach speeds of over 100 miles per hour (160 km/hour).

Debris flows occur after wildfires have burned the vegetation from a steep slope. Before the fire the vegetation's roots anchored the soil on the slope and removed water from the soil. The loss of support and accumulation of moisture can result in a catastrophic failure. Rainfall that was previously absorbed by vegetation now runs off immediately. A moderate amount of rain on a burn scar can trigger a large debris flow.

Rescuers assigned to a Mud and Debris flow Incident should take steps to make sure that the following guidelines are followed:

A. First Responders need to consider Apparatus and Personnel placement when responding to a report of a Mud and Debris flow incident since "*Waters Edge"* can be difficult to recognize. Historically, these incidents occur in multiple waves. First Responders need to recognize the need to post lookouts upstream.

B. All personnel assigned to the incident should make sure that they have donned their appropriate PPE prior to arrival at the incident.

C. First Responders need to consider the *Risk vs. Benefit* of removing civilians or committing resources to the affected area. Nighttime operations involving these incidents dramatically increase the risk to First Responders.

APPENDIX C

PERSONAL SAFETY EQUIPMENT

The safety of personnel is the most important factor in any rescue operation. This is especially true when there is a risk of rescuers falling into moving water. Due to this hazard, Specialized Personal Safety Equipment is mandatory for rescuers in two cases:

1. In any position where an inadvertent action may cause a member to fall into the water, (i.e., sloped banks of tapered channels)

2. In all cases where a member is *"Within Ten (10) feet of the Water's Edge".*

SAFETY EQUIPMENT

Absolutely No Turnout Clothing shall be worn on these types of calls, including Rubber Boots and Structural Firefighting Helmets. The following are considered acceptable Personal Safety Equipment:

A. PERSONAL FLOTATION DEVICE (PFD):Coast Guard Approved Type III, Type V, or Type III/ V Personal Flotation Devices (PFD), are recommended. These types of PFD's provide a minimum of 15 1/2 pounds of flotation. The fit is the most important feature when selecting a PFD. Try to find a Jacket that provides the most flotation while fitting you properly.

B. HELMETS: Light, ventilated helmets (self-draining), without a fixed brim, but with foam padding or a suspension system, are the preferred types.

C. KNIFE: A good knife is essential in SF/S&R Incidents. This knife is considered a lifesaving device and should not be used for any other purpose. The knife should be straight-bladed and attached to the shoulder of the PFD, handle down. A sheath that keeps the knife in place, and blade sharpness are the most important considerations.

D. WHISTLE: A good whistle is necessary for communication. It should be loud and work when wet. The whistle is attached to the shoulder of the PFD.

E. GLOVES: A combination of Neoprene and Leather Gloves provide the best blend of thermal protection and manual dexterity, although all-leather gloves will work.

APPENDIX C (Continued)

F. FOOTWEAR: Tennis shoes with wool socks or Neoprene socks are a good combination. Neoprene booties with thick soles are another good choice. Leather boots are acceptable in a supportive position away from the water's edge.

THERMAL PROTECTION

A. FIRST RESPONDERS: Lightweight Wildland Gear including pants, jackets, gloves, and or light Raingear are acceptable Safety Equipment.

B. SWIFTWATER/FLOOD SEARCH AND RESCUE TEAMS: Wetsuits are an excellent choice when cost is a consideration. Dry suits provide the best thermal protection, ease of movement, fit different sized people, provide wind protection when out of the water, and provide hazardous material protection when working in contaminated water.

APPENDIX D

RECOGNIZED SWIFTWATER/FLOOD - SEARCH AND RESCUE TEAMS

Agency / Resource Designator	SF/S&R Capability / Type	Notes
Anaheim Fire Department – Engine 2 & US&R 2 (ANA E2 & US&R2)	Type 2 – Modified	Modified - (Helicopter Awareness Only)
Orange Fire Department – US&R1 (ORG US&R1)	Type 3 – Modified	Modified - (No Non-Power IRB Capability)
Orange County Fire Authority – Swift Water 6 (ORC SW6)	Type 3	
Orange County Fire Authority – Swift Water 32 (ORC SW32)	Type 3	
Orange County Fire Authority – Swift Water 56 (ORC SW56)	Type 3	
Orange County Fire Authority – US&R Support 61 (ORC SW61)	Type 3	
Orange County Fire Authority (OES SW5)	Type 1	Requires call-back staffing of 14 personnel

*Typing was taken from FIRESCOPE ICS-SF/SAR-020-1, *"Operational System Description"*, page 10, Appendix A – Swiftwater/Flood Search and Rescue Typing

Swiftwater/Flood Search and Rescue Resource Typing

	Type 1	Type 2	Type 3	Type 4
SF/SAR Type (Capabilities)	Manage search ops Power vessel ops In-water contact rescues Helicopter operational Technical rope systems HazMat Animal rescue EMS-ALS Communications Logistics Capable of 24hr operations	Manage search ops Power vessel ops In-water contact rescues Helicopter operational Technical rope systems HazMat Animal rescue EMS-BLS Capable of 24hr operations	In-water contact rescues Assist in search ops Non-power watercraft HazMat Animal rescue EMS-BLS Capable of 24hr operations	Low Risk Land Based HazMat EMS-BLS Capable of 24hr operations

FIRESCOPE, SWIFTWATER/FLOOD SEARCH AND RESCUE, *"Operational System Description",* ICS US&R 120-2 & Law Enforcement Mutual Aid Plan (SAR) Annex, (January 24, 2001)

APPENDIX D (Continued)
(SF/S&R Team Training)

It is recommended that teams have the following skills. A number indicates the minimum number of members on the team with indicated skills. "X" indicates recommended skills for all team members.

	Swiftwater/Flood Search & Rescue Team				**FEB**
	Type 1	Type 2	Type 3	Type 4	
Animal Rescue	2	1	1		1
Class 3 Paddle Skills	X	X	X		
Communications Skill (Radios)	1				
Contact and Self Rescue Skills	X	X	X		
EMS - ALS Capability	2				
EMS - BLS Capability	X	X	X	X	X
Hazardous Materials	X	X	X	X	X
Helicopter/Aquatic Rescue Operational Skills	4	2			
Helicopter Operations Awareness	X	X			
Incident Command System (ICS)	X	X	X	X	X
Powered Boat Operator	4	2			2
SF/SAR Management	X	X			
SF/SAR Operations	X	X	X	X	X
Technical Rope Rescue Skills	X	X			
SCUBA Support	4*				

***= Optional**

APPENDIX D (Continued) SF/S&R Team Training

It is recommended that team personnel have completed the following training as indicated. A number indicates the minimum number of person(s) on a team with the indicated training. "X" indicates recommended training for all team members.

		SF/S&R Type 1 Team Manager	Swiftwater/Flood Search & Rescue Team				FEB
			Type 1	Type 2	Type 3	Type 4	
1	Annual skills proficiency demonstration (by agency)	X	X	X	X	X	
2	Annual swim demonstration (by agency)	X	X	X	X	X	
3	EMS – Advanced First Aid & CPR			X	X	X	X
4	EMS – Emergency Medical Technician			X	1	1	
5	EMS – Paramedics/ALS			2			
6	Hazardous Materials	X	X	X	X	X	X
7	Helicopter Operations Awareness*	X	X	X			
8	Helicopter/Aquatic Rescue or Equivalent*		4	2			
9	Incident Command System (ICS)	X	X	X	X	X	X
10	Power Craft Operators Training* (per type of craft on team)		4	2			2
11	Swiftwater/Flood Rescue Operational					X	
12	Swiftwater/Flood Rescue Technician**	X	X	X	X		
13	Rope Rescue Technician**	X	X	X			
14	Search Management	X	X	X			
15	Technical Animal Rescue or Equivalent		2	1	1	1	1
16	Basic SCUBA (with equipment orientations recommended)		4				

= Annual refresher** *= Three (3) year refresher**